Kartavya Sharma

Austin, TX, 78660 • 512.783.4540 • Kartavya.Sharma@hotmail.com • LinkedIn • Portfolio

PROFILE

Motivated Computer Science Master's student specializing in high-performance parallel systems and Generative AI (RAG) architectures. Proven expertise in engineering scalable embedding pipelines and cloud-native workflows on Google Cloud Platform to optimize retrieval. Proficient in Python and Java, utilizing LangChain to architect production-ready solutions. Leverages Agile methodologies to lead software initiatives within collaborative, team driven environments.

SUMMARY OF QUALIFICATIONS

- Proficient in programming and scripting with Python, Java, C/C++, and x86 Assembly, applying these skills in parallel computing, multiprocessing, and the performance optimization of complex, high-performance simulation systems.
- Experienced in Generative AI, Retrieval-Augmented Generation (RAG), and vector embeddings, leveraging Large Language Models and semantic search to enhance information retrieval, system intelligence, and contextual decision-making.
- Skilled in cloud-native data engineering and system development with Google Cloud Platform, BigQuery, and LangChain, integrating automated ELT/ETL workflows with AI architectures for end-to-end intelligent solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C/C++, JavaScript, HTML/CSS, x86 Assembly, Linux (Environment/Bash).
- **Systems/Libraries:** LangChain, HuggingFace, ChromaDB, Groq, PyTorch, NumPy, Pandas, Matplotlib, ReactJS, Git/GitHub.
- **Areas of Expertise:** Generative AI (RAG), Large Language Models (LLMs), Vector Embeddings & Search, Cloud-Native AI (GCP), Natural Language Processing (NLP), Neural Networks, Machine Learning, and Parallel Computing.

EDUCATION

Bachelor of Science in Computer Science - *Texas Tech University, Lubbock, TX* Fall 2027

- **GPA:** 3.5
- **Awards:** Dean's List Spring 2025, Fall 2025

Accelerated Master of Science in Computer Science – *Texas Tech University, Lubbock, TX* Fall 2027

- **GPA:** 4.0

PROJECTS

Parallel Cellular Automaton Simulator (Python, Multiprocessing, Argparse) – *Concepts of Programming Languages* Dec 2025
Lubbock, TX

- Engineered a high-performance simulation engine in Python using multiprocessing to parallelize row-based computations, significantly reducing runtime for 100-step evolution cycles on large grid matrices.
- Designed a complex cellular evolution based on number theory concepts (Prime, Fibonacci, Power of 2) within a wrapped toroidal grid to generate dynamic simulation patterns
- Built a robust command line interface with Argparse, incorporating input validation, scalable process control, and consistent file I/O; validated with correctness by cross testing parallel and serial outputs.

Generative AI RAG BOT (Python, LangChain, Groq, LLMs, ChromaDB, HuggingFace) – *Independent Project* Oct 2024
Lubbock, TX

- Developed a Retrieval-Augmented Generation (RAG) chatbot integrating LangChain, Groq LLMs, and ChromaDB to enable context aware question answering over custom datasets.
- Optimized retrieval and inference pipelines by refining text chunking and vector search flow, reducing latency and improving response relevance.
- Architected a modular backend with decoupled loaders, retrievers, and prompt pipelines, improving scalability, maintainability, and grounded response accuracy.

ACTIVITIES / CERTIFICATIONS

Google Cloud Skills AI Boost Certificate – *Google Cloud Platform* Jul 2024

- Completed advanced training in gen AI, covering LLM fundamentals, prompt engineering, and responsible AI concepts.

Microsoft Excel & Word Certification – *Microsoft Office Specialist* May 2023

- Applied advanced data modeling and business intelligence to design automated reports and interactive dashboards.

Member – *Google Developer Group, TTU, Lubbock TX* Oct 2023 - Present

Member – *Indian Student Association, TTU, Lubbock TX* Aug 2023 - Present